

Md. Rifat Mahmud

Junior AI Engineer • Full-Stack & Applied AI

+880-1936796230 • Email • LinkedIn • GitHub • Portfolio • Dhaka, Bangladesh

Professional Summary

Junior AI Engineer at Impala Intech and CSE graduate (ULAB, CGPA 3.55/4.00) building full-stack web products and applied AI across NLP, chatbots, RAG, and document processing. Builds backends and admin panels (Node.js, NestJS, FastAPI, Next.js) and Flutter mobile apps. First-author paper on explainable deep learning for chest X-ray diagnosis accepted at IEEE CSDE 2026. Strong in Python, TypeScript, and Java, with a foundation in OOP, data structures, and algorithms.

Experience

- **Impala Intech** Dhaka, Bangladesh (on-site)
Junior AI Engineer Jun 2026 – present
 - Develop and deploy AI-powered products full-time: NLP, chatbots, and document processing.
 - Promoted to a permanent role after completing the internship on the same team.
- **Impala Intech** Dhaka, Bangladesh (on-site)
AI Engineer Intern Feb 2026 – May 2026
 - Four-month internship on NLP, chatbot development, and document processing.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, Java, PHP, C, Dart, SQL
- **ML / AI:** PyTorch, TensorFlow, Keras, OpenCV; Grad-CAM++, LIME, SHAP
- **LLM / NLP:** LangChain, RAG, Ollama, ChromaDB, sentence-transformers
- **Backend & data:** Node.js, Express, NestJS, FastAPI, Flask, Prisma, PostgreSQL, MySQL, Redis, WebSockets, Socket.IO, BullMQ
- **Frontend & mobile:** Next.js, React, Tailwind CSS, Flutter
- **DevOps & tools:** Docker, Git/GitHub, Vitest, LaTeX
- **Foundations:** OOP, Data Structures, Algorithms, Clean Architecture
- **Spoken languages:** Bangla (native), English (intermediate)

Education

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
B.Sc. Computer Science and Engineering Jun 2022 – Summer 2026
CGPA: 3.55/4.00. Thesis: *Explainable Hybrid Ensemble Diagnosis of Five Lung Conditions from Lung-Segmented Chest X-Rays* (Supervisor: Prof. Muhammad Golam Kibria).

Research & Technical Projects

Gavirad Mart — Points & Rewards Backend and Admin Panel 2026

- Backend and admin panel for a points and rewards app for pharma field sales officers; daily sales turn into points via targets, bonus bands, and flash multipliers.
- Clean-architecture API built for 10,000 concurrent users: ledger balances, row locks, idempotent redeems, SMS OTP login, and live stock over WebSocket.
- Node.js, Express 5, TypeScript, Prisma, MySQL, WebSocket, Next.js 16, React 19.

Explainable Hybrid Ensemble Diagnosis of Five Lung Conditions ULAB thesis, 2026

- Five-class chest X-ray system (bacterial/viral pneumonia, COVID-19, tuberculosis, normal) that segments the lungs and refuses a label when the upload fails the lung check.
- Ensemble of DenseNet201, ResNet50, VGG16, and EfficientNet-B3 fine-tuned on 35,686 lung-cropped images: 89.23% accuracy and 91.67% macro-F1 on 7,651 held-out images.
- Grad-CAM++, LIME, and SHAP explanations constrained to the lung mask.

- Python, PyTorch, OpenCV, PSPNet, CLAHE, Grad-CAM++, LIME, SHAP, Flask.

Radiant Nutraceuticals Pharma Assistant — AI Chatbot

- Bilingual English/Bangla chatbot for 33 herbal products: dosage, side effects, warnings, and interactions; RAG over the product catalog.
- Admin pipeline extracts PDF, PPTX, DOCX, and TXT to refresh the medicine database.
- Python, FastAPI, Ollama, ChromaDB, sentence-transformers.

YNNotes — Academic Notes Marketplace

- Three-role EdTech marketplace (Student, Creator, Admin) with 71+ REST endpoints over a normalized PostgreSQL schema.
- SSLCommerz wallet top-ups (bKash, Nagad, Rocket), secure PDF delivery via Cloudflare R2, and a 70/30 creator revenue share.
- Next.js 14, TypeScript, PostgreSQL, Prisma, NextAuth v5, Tailwind, Cloudflare R2, Redis.

Easy Prescription

- Scans a handwritten prescription, matches medicines with Levenshtein distance and Soundex, then saves, speaks, or exports it as a PDF.
- English and Bangla UI, OTP login, and history; same flow on the web and in a Flutter app.
- React, TypeScript, Express, MySQL, Flutter, OCR.space, ElevenLabs.

Also: FIFA World Cup Prediction App; Brand Lifecycle Task Manager; Novara . Full write-ups on my **portfolio**.

Publications

- **Mahmud, M.R., Maharab, S.M.A., Akter, T., Ansari, S., & Kibria, M.G.:** *Multi-class Lung Disease Diagnosis Using Chest X-Ray with XAI*. Accepted at the 11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (IEEE CSDE 2026), Cox's Bazar, Bangladesh.

Honors & Activities

- Participant: UCPC Take-Off Programming Contest (Fall 2023), solved 4 of 5 problems

References

Prof. Muhammad Golam Kibria, PhD, SMIEEE

Professor & Head, Department of CSE, University of Liberal Arts Bangladesh (ULAB)

Email: golam.kibria@ulab.edu.bd